UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	INTRODUCTION TO INTERACTIVE MULTIMEDIA
COURSE CODE	CSC413
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (15 MARKS)

State TRUE (T) or FALSE (F) for the following statements.

1. A web browser displays hypertext documents.
2. Authoring tools help developers organise and edit multimedia elements.
3. A scanner is an input device.
4. Hypermedia combines multimedia with hyperlinks.
5. A vector image becomes pixelated when enlarged.
6. A storyboard is part of project planning.
7. Interactivity gives users control over content and flow.
8. 3D animation does not use modelling.
9. MIDI files store digitized audio samples.
10. A storyboard is a visual plan of a multimedia sequence.
11. USB is a plug-and-play connection standard.
12. Interactivity level is irrelevant when choosing an authoring tool.
13. A plotter is an input device.
14. A prototype is an early model of a product.
15. A multimedia project always uses every media element.

PART B (15 MARKS)

Select the most appropriate answer from Table A for each of the following questions.

Table A

A. Bitmap	B. Virtual reality	C. Keyframe	D. Tweening	E. Sampling
F. Frame rate	G. Hypermedia	H. Multimedia	I. Linear multimedia	J. Chroma key
K. Path animation	L. Vector	M. Cel animation	N. MIDI	O. Browser

1. integration of text, graphics, audio, video and animation
2. important frame in an animation
3. converting sound into digital values
4. generating in-between frames
5. presentation without user control of sequence
6. hand-drawn frame-by-frame animation
7. linked multimedia elements
8. moving an object along a route
9. standard for musical notes and instructions
10. simulated environment
11. frames displayed per second
12. replacing a colour background
13. image made of pixels
14. software that displays web pages
15. image made of mathematical objects

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PART C (30 MARKS)

Choose the correct answer for each question.

1. Anti-aliasing is used to:
A. Compress video
B. Smooth jagged edges
C. Add sound
D. Increase file size only

2. Which is NOT a multimedia element?
A. Animation
B. Audio
C. Compiler
D. Text

3. Interlacing:
A. Removes colour
B. Adds sound
C. Draws alternate lines of a frame
D. Draws every line at once

4. Resolution refers to:
A. Number of pixels in an image
B. File type
C. Font size
D. Colour name

5. MPEG is:
A. A video compression family
B. An image editor
C. A storage device
D. A font

6. Card-based authoring tools are suited to:
A. Audio editing only
B. Live video only
C. Messages with a beginning and end
D. 3D modelling only

7. Which is an audio file format?
A. JPEG
B. WAV
C. MPEG
D. HTML

8. A licence gives:
A. Permission to use a work
B. Unlimited bandwidth
C. Ownership of all works
D. Free hardware

9. Hertz measures:
A. Frequency
B. File size
C. Loudness
D. Resolution

10. Which is an authoring tool category?
A. CMYK
B. Card-based
C. JPEG
D. RGB

11. A graphics tablet is used for:
A. Drawing input
B. Web browsing
C. Printing
D. Sound output

12. A storyboard is:
A. A colour model
B. A visual sequence plan
C. A sound file
D. A video codec

13. Streaming allows media to:
A. Require full download first
B. Play while downloading
C. Remove compression
D. Only play offline

14. Amplitude of a sound wave relates to:
A. Pitch
B. Colour
C. Frame rate
D. Loudness

15. Which best defines multimedia?
A. A single image
B. A printed book
C. Text only
D. Text, graphics, audio, video and animation

16. A hyperlink links to:
A. Only text
B. Only audio
C. Only images
D. Another resource

17. RGB is used mainly for:

A. Video editing only
B. Audio recording
C. Screen display
D. Printed output

18. Plug and play means:

A. No power needed
B. Manual driver installation only
C. Wireless only
D. Automatic device recognition

19. CMYK is used mainly for:

A. Animation
B. Sound mixing
C. Printed output
D. Screen display

20. Cel animation is created by:

A. Recording sound
B. Scanning text
C. Printing images
D. Drawing each frame by hand

21. Which is an input device?

A. Plotter
B. Monitor
C. Printer
D. Scanner

22. Frame rate is:

A. Bits per second
B. Frames per second
C. Samples per second
D. Pixels per inch

23. A bitmap image is made of:

A. Mathematical objects
B. Pixels
C. Text characters
D. Sound samples

24. A browser is used to:

A. Scan images
B. Edit audio
C. Display web pages
D. Print documents only

25. Path animation moves an object:

A. Without motion
B. Only vertically
C. Randomly
D. Along a predefined route

26. Kerning adjusts:

A. Page size
B. Font colour
C. Line length
D. Space between individual character pairs

27. Convergence in multimedia refers to:

A. Recording audio only
B. Printing images
C. Deleting media files
D. Merging of media and computing

28. Which is a benefit of multimedia in education?

A. It replaces all teachers
B. It combines visuals and sound
C. It uses only text
D. It removes all interaction

29. A serif font has:

A. No strokes
B. No spacing
C. Small strokes at character ends
D. Only uppercase letters

30. Hypertext contains:

A. Links to other content
B. No text
C. Only audio
D. Only images

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PART D (40 MARKS)

QUESTION 1

A product catalogue must show sharp images on screen and in print.

a) State TWO differences between bitmap and vector images.

(2 marks)

b) Describe TWO factors that affect image quality.

(4 marks)

c) Explain how colour model and resolution affect screen and print output.

(4 marks)

QUESTION 2

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

(2 marks)

b) Describe TWO roles of hyperlinks in the website.

(3 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

(5 marks)

QUESTION 3

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

(3 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

(3 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

(4 marks)

QUESTION 4

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

(3 marks)

b) Explain the interactivity level suitable for the demonstration.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

(4 marks)

END OF QUESTION PAPER